

R Programming Tutorial

Introduction to R Programming

What is R?

R is an open-source programming language and software environment used for:

- Statistical computing
- Data analysis
- Data visualization

It is widely used by statisticians, data scientists, and researchers.

History of R

- Developed in the 1990s by Ross Ihaka and Robert Gentleman
- Created at the University of Auckland, New Zealand
- Inspired by the S programming language

Features of R

- Free and open-source
- Cross-platform (Windows, Linux, macOS)
- Powerful data handling and storage
- Large collection of packages
- Excellent graphical capabilities

Applications of R

- Data analysis and visualization
- Machine learning
- Statistical modeling
- Financial analysis
- Bioinformatics
- Research and academia

Advantages of R

- Free to use
- Strong community support
- Thousands of packages available

- Ideal for statistical analysis

Limitations of R

- Slower than some languages (e.g., C++)
- Uses more memory
- Not ideal for large-scale production systems

1. Install R

- Go to the official CRAN site: <https://cran.r-project.org>
- Click “**Download R for Windows**” → “**base**” → “**Download R for Windows**”
- Run the installer (.exe) and follow default settings (safe for beginners)

✓ This installs the core R engine.

2. Install RStudio (Recommended)

R alone is basic; RStudio makes it much easier to work with.

- Visit: <https://posit.co/download/rstudio-desktop/>
- Download **RStudio Desktop (free version)**
- Install it normally

✓ RStudio automatically detects your R installation.

3. Verify Installation

- Open **RStudio**
- In the console (bottom-left), type:

```
print("Hello, R!")
```

If it prints correctly, you're set.

4. Install Essential Packages

In RStudio console, run:

```
install.packages("tidyverse")
install.packages("data.table")
install.packages("ggplot2")
```

These help with:

- Data manipulation
 - Fast processing
 - Visualization
-

5. (Optional) Set Working Directory

This tells R where your files are stored:

```
setwd("C:/Users/YourName/Documents")
```

6. (Optional) Add R to PATH (Advanced)

Usually not needed, but if you want to run R from Command Prompt:

- Search **Environment Variables**
- Edit Path
- Add something like:

```
C:\Program Files\R\R-4.x.x\bin
```

Common Issues (and fixes)

- **RStudio not finding R** → Reinstall R first, then RStudio
- **Package install errors** → Run RStudio as Administrator
- **Internet issue** → Choose a CRAN mirror manually:

```
chooseCRANmirror()
```

Quick Tip

If you're just starting, focus on learning:

- Vectors & data frames
- `dplyr` (data manipulation)

- `ggplot2` (visualization)

Data Types and Variables in R

1. Variables in R

A variable is a name used to store a value (data) in memory.

Rules for Naming Variables

- Must start with a letter or `.`
- Cannot start with a number
- No spaces allowed
- Case-sensitive (`ⓧ` and `x` are different)

Assigning Values

R uses `<-` (preferred) or `=`:

```
x <- 10
name <- "Ali"
y = 5
```

2. Data Types in R

Data types define the kind of value a variable can store.

Main Data Types

1. Numeric

- Represents numbers (integers & decimals)

```
x <- 10
y <- 3.14
```

2. Character

- Represents text (strings)

```
name <- "R Programming"
```

3. Logical

- Represents boolean values

```
a <- TRUE  
b <- FALSE
```

4. Integer

- Whole numbers (use `L`)

```
num <- 5L
```

5. Complex

- Numbers with real and imaginary parts

```
z <- 2 + 3i
```

3. Checking Data Types

Use built-in functions:

```
typeof(x)  
class(x)
```

4. Type Conversion (Casting)

Convert one data type to another:

```
as.numeric("10")  
as.character(10)  
as.logical(1)
```

5. Special Values in R

- `NA` → Missing value
- `NULL` → Empty value
- `Inf` → Infinity
- `NaN` → Not a number

```
x <- NA  
y <- Inf  
z <- NaN
```

6. Constants vs Variables

- **Variable** → Value can change
 - **Constant** → Fixed value (not formally defined in R, but treated as unchanged)
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7. Examples

```
age <- 20  
height <- 5.8  
is_student <- TRUE  
name <- "Ali"
```

```
print(age)  
print(name)
```